Composition Series

Will Gilroy

July 2026

As of $18th$ July this note is still in rough form

Let R be a local ring with maximal ideal $\mathfrak{m} = (y_1, \cdots, y_m)$ and let $M = R/I$ be an R-module where I is $\mathfrak{m}$-primary. We want to try and construct a composition series for M by iteratively constructing a chain where the partial quotient is R/I' where $I' = (\text{gens } I, y_1)$. Let's try to work out how this might work with an example

Let $R = k[x]_{(x)}$ and $M = R/(x^3)$. The correspondence theorem gives us that the submodules of M correspond to the ideals $0, (x^2), (x^3)$, giving the submodules $0/(x^3), (x^2)/(x^3)$, and $(x^3)/(x^3)$.

Mike was giving us a procedure so that the first partial quotient would be something like $R/(x^3, x) = R/(x)$ which is now modding out by the maximal ideal and the procedure would terminate. I think we would then get the wrong length by this procedure, but I admit I do not totally recall the initial chain that we would start with. Perhaps we can try this example again with $R = k[x, y]_{(x,y)}$ and $R/(x^2, y^2)$?

.1 Correspondence Theorem and Submodules of R/I.

A quick comment on how the correspondence theorem gives us the submodules of $R/(x^3)$. Recall the correspondence theorem for groups: if G is a group and $N \leq G$ a normal subgroup then there is a bijection

$$\{\text{Subgroups of } G \text{ containing } N\} \leftrightarrow \{\text{Subgroups of } G/N\}.$$

This correspondence theorem extends to R modules in the following way. Let R be a ring, viewed as an R module, and let N an submodule of R. Then the quotient R-module R/N is in particular a quotient group of R. Applying the correspondence theorem for groups gives us that the subgroups (and thus the only contenders for the submodules of R/N) of R/N are $\{R/L : N \subset L \leq R\}$. We can extend all of these subgroups to R submodules of R/N in the usual way and we thus have the following bijection

$$\{\text{Submodules of } R \text{ containing } N\} \leftrightarrow \{\text{Submodules of } R/N\}.$$

In particular, if $M = R/I$ where $I \leq R$ is an ideal of R applying this bijection gives

$$\{\text{Ideals of } R \text{ containing } I\} \leftrightarrow \{\text{Submodules of } R/I\},$$

noting that the submodules of R are exactly its ideals.

Example 1. Let's apply this to the example in the main body above, let $R = k[x]$ and $M = k[x]/(x^3)$. To construct a chain of M we need to understand its submodules. The submodules of M are in bijection with the ideals of R containing (x^3). These ideals are exactly $(1), (x), (x^2), (x^3)$. Applying the bijection map (which we did not describe) gives the submodules

$$R/(x^3), (x)/(x^3), (x^2)/(x^3), (x^3)/(x^3),$$

where each of these quotients are the quotient modules (note for example that the ideal (x) is a subgroup of (x^3) when viewed purely as additive groups.) we can then construct a composition series

$$(x^3)/(x^3) = 0 \subseteq (x^2)/(x^3) \subseteq (x)/(x^3) \subseteq R/(x^3) = (1)/(x^3).$$

Since we have enumerated all of the submodules of $R/(x^3)$ we can see that there is no longer chain which can be constructed (in particular, we have a chain which has used all of the submodules[1]) and so the above chain is in fact a composition series (one can also check that the partial quotients are simple if one prefers). Thus we have shown that the length

$$\ell_R(R/(x^3)) = 3.$$

Note that following a similar argument shows that $\ell_R(R/(x^n)) = n$. I think we can also further note that, since localisation is an exact functor, the above chain remains a chain and so the length should not change under localising at (x). We should work out the details for this.

[1] In this case it turns out that we have a total order under inclusion of the submodules of M. In general one does not need a condition this strong: it is sufficient to have some maximal chain of submodules with respect to inclusion.